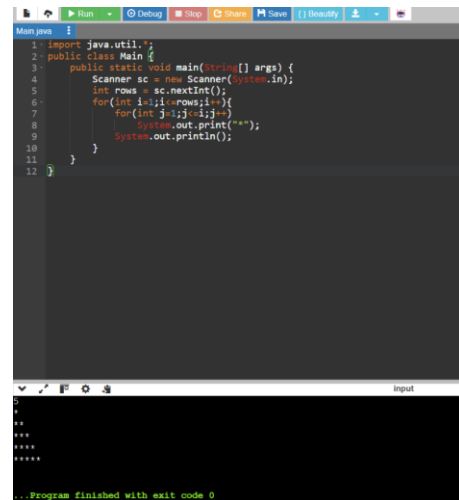


NAME: G VISHNU MADHAV

REG NO. 192425192

1. Right-Angled Triangle of Stars



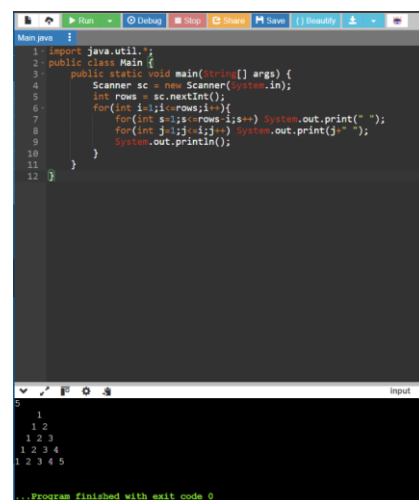
```
1: import java.util.*;
2: public class Main {
3:     public static void main(String[] args) {
4:         Scanner sc = new Scanner(System.in);
5:         int rows = sc.nextInt();
6:         for(int i=1;i<=rows;i++){
7:             for(int j=1;j<=i;j++){
8:                 System.out.print("*");
9:                 System.out.println();
10:            }
11:        }
12:    }
}
```

Input

```
5
*
**
***
****
*****

...Program finished with exit code 0
```

2: Number Pyramid

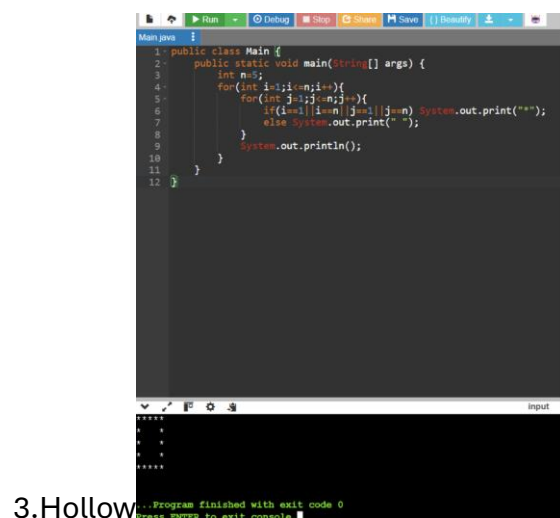


```
1: import java.util.*;
2: public class Main {
3:     public static void main(String[] args) {
4:         Scanner sc = new Scanner(System.in);
5:         int rows = sc.nextInt();
6:         for(int i=1;i<=rows;i++){
7:             for(int s=1;s<=rows-i;s++) System.out.print(" ");
8:             for(int j=1;j<=i;j++) System.out.print(j+" ");
9:             System.out.println();
10:        }
11:    }
12: }
```

Input

```
5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

...Program finished with exit code 0
```



```
1: public class Main {
2:     public static void main(String[] args) {
3:         int n=5;
4:         for(int i=1;i<=n;i++){
5:             for(int j=1;j<=n;j++){
6:                 if(i==1||i==n||j==1||j==n) System.out.print("*");
7:                 else System.out.print(" ");
8:             }
9:             System.out.println();
10:        }
11:    }
12: }
```

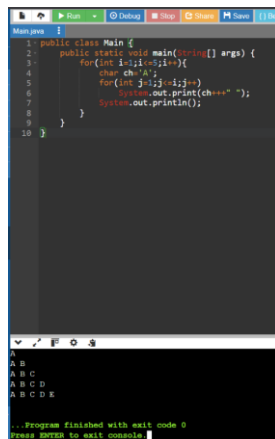
Input

```
*****
*   *
*   *
*   *
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

3. Hollow Square

4. Alphabet Triangle



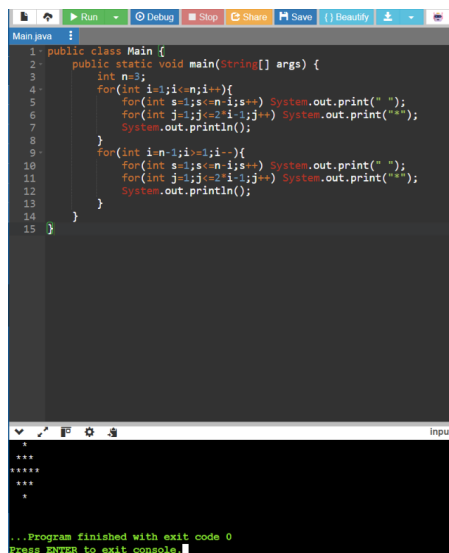
```
1: public class Main {
2:     public static void main(String[] args) {
3:         for(int i=1;i<=5;i++){
4:             char ch='A';
5:             for(int j=1;j<=i;j++){
6:                 System.out.print(ch+" ");
7:                 System.out.println();
8:             }
9:         }
10:     }
}
```

Output:

```
A B
A B C
A B C D
A B C D E
```

...Program finished with exit code 0
Press ENTER to exit console.

5. Diamond Pattern



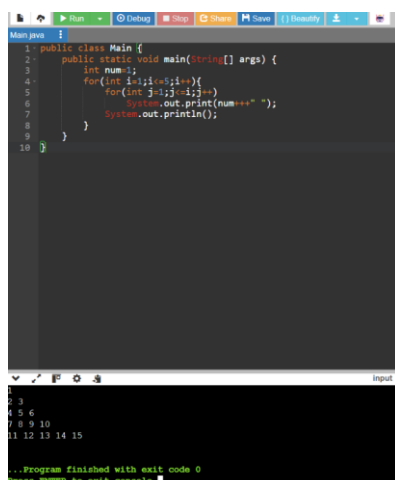
```
1: public class Main {
2:     public static void main(String[] args) {
3:         int n=5;
4:         for(int i=1;i<=n;i++){
5:             for(int s=1;s<=n-i;s++) System.out.print(" ");
6:             for(int j=1;j<=2*i-1;j++) System.out.print("*");
7:             System.out.println();
8:         }
9:         for(int i=n-1;i>=1;i--){
10:             for(int s=1;s<=n-i;s++) System.out.print(" ");
11:             for(int j=1;j<=2*i-1;j++) System.out.print("*");
12:             System.out.println();
13:         }
14:     }
15: }
```

Output:

```

*
***
*****
***
*
...Program finished with exit code 0
Press ENTER to exit console.
```

6. Floyd's Triangle



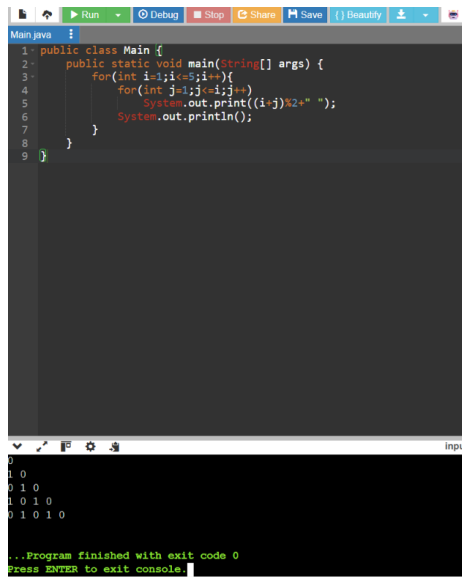
```
1: public class Main {
2:     public static void main(String[] args) {
3:         int num=1;
4:         for(int i=1;i<=5;i++){
5:             for(int j=1;j<=i;j++){
6:                 System.out.print(num+" ");
7:                 num++;
8:             }
9:             System.out.println();
10:         }
}
```

Output:

```
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
```

...Program finished with exit code 0
Press ENTER to exit console.

7. Binary Triangle



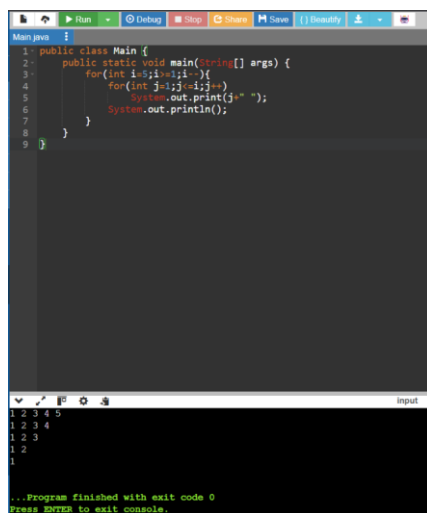
```
1: public class Main {
2:     public static void main(String[] args) {
3:         for(int i=1; i<=5; i++){
4:             for(int j=1; j<=5; j++){
5:                 System.out.print((i+j)%2+" ");
6:             }
7:             System.out.println();
8:         }
9:     }
}
```

Input

```
0
1 0
0 1 0
1 0 1 0
0 1 0 1 0

...Program finished with exit code 0
Press ENTER to exit console.
```

8. Inverted Number Triangle



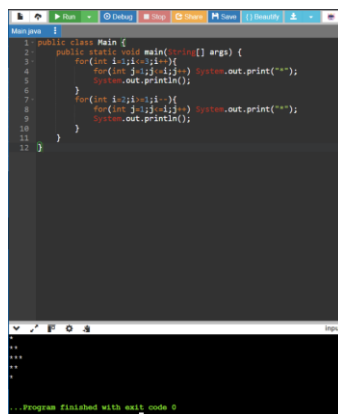
```
1: public class Main {
2:     public static void main(String[] args) {
3:         for(int i=5; i>=1; i--){
4:             for(int j=i; j>=1; j--){
5:                 System.out.print(j+" ");
6:             }
7:             System.out.println();
8:         }
9:     }
}
```

Input

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

...Program finished with exit code 0
Press ENTER to exit console.
```

9. Half Diamond



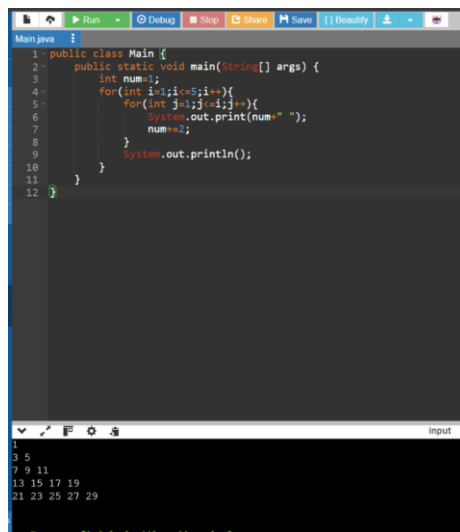
```
1: public class Main {
2:     public static void main(String[] args) {
3:         for(int i=1; i<=5; i++){
4:             for(int j=1; j<=i; j++){
5:                 System.out.print(" ");
6:             }
7:             for(int k=i; k>=1; k--){
8:                 for(int l=1; l<=i; l++){
9:                     System.out.print(" ");
10:                 }
11:             }
12:         }
13:     }
}
```

Input

```
..
**
***
****
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

10. Odd Number Triangle



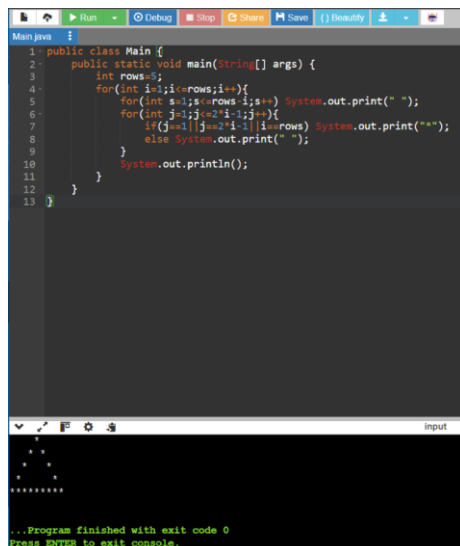
```
1 public class Main {
2     public static void main(String[] args) {
3         int num=1;
4         for(int i=1;i<=5;i++){
5             for(int j=1;j<=i;j++){
6                 System.out.print(num+" ");
7                 num+=2;
8             }
9             System.out.println();
10        }
11    }
12 }
```

Output:

```
1
3 5
7 9 11
13 15 17 19
21 23 25 27 29
```

Program finished with exit code 0

11. Hollow Pyramid



```
1 public class Main {
2     public static void main(String[] args) {
3         int rows=5;
4         for(int i=1;i<=rows;i++){
5             for(int s=1;s<=rows-i;s++) System.out.print(" ");
6             for(int j=1;j<=2*i-1;j++){
7                 if((j==1||j==2*i-1||i==rows) System.out.print("*");
8                 else System.out.print(" ");
9             }
10            System.out.println();
11        }
12    }
13 }
```

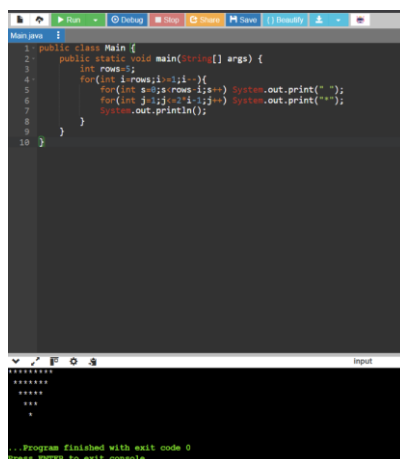
Output:

```

 *
 * *
* * *
* * * *
*****
```

...Program finished with exit code 0
Press ENTER to exit console.

12. Reverse Pyramid



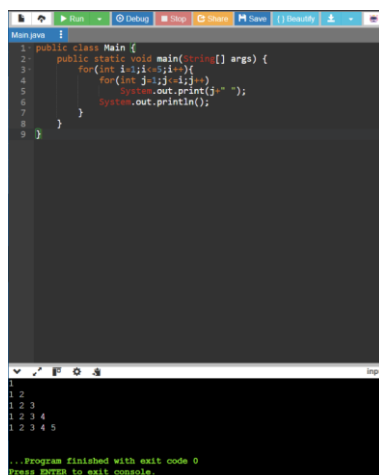
```
1 public class Main {
2     public static void main(String[] args) {
3         int rows=5;
4         for(int i=rows;i>=1;i--){
5             for(int s=0;s<=rows-i;s++) System.out.print(" ");
6             for(int j=i;j>=1;j--) System.out.print("*");
7             System.out.println();
8         }
9     }
10 }
```

Output:

```
*****
****
***
**
*
```

...Program finished with exit code 0
Press ENTER to exit console.

13. Number Right-Angled Triangle

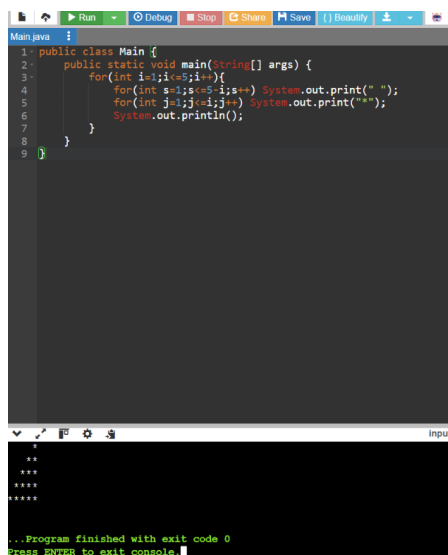


```
1 public class Main {
2     public static void main(String[] args) {
3         for(int i=1;i<=5;i++){
4             for(int j=1;j<=i;j++){
5                 System.out.print(j+" ");
6             }
7             System.out.println();
8         }
9     }
10 }
```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

.. Program finished with exit code 0
Press ENTER to exit console.

14. Right-Aligned Star Triangle

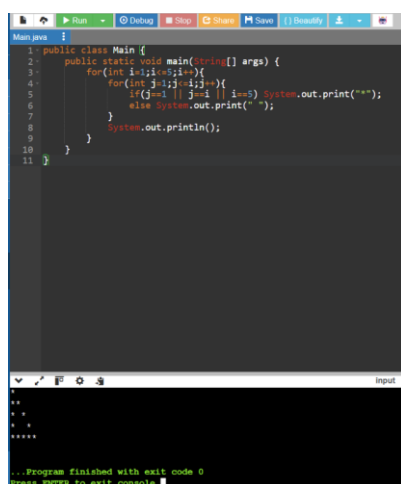


```
1 public class Main {
2     public static void main(String[] args) {
3         for(int i=1;i<=5;i++){
4             for(int m=1;m<=5-i;m++) System.out.print(" ");
5             for(int j=1;j<=i;j++) System.out.print("*");
6             System.out.println();
7         }
8     }
9 }
```

1
*
*
*
*

.. Program finished with exit code 0
Press ENTER to exit console.

15. Hollow Right-Angled Triangle

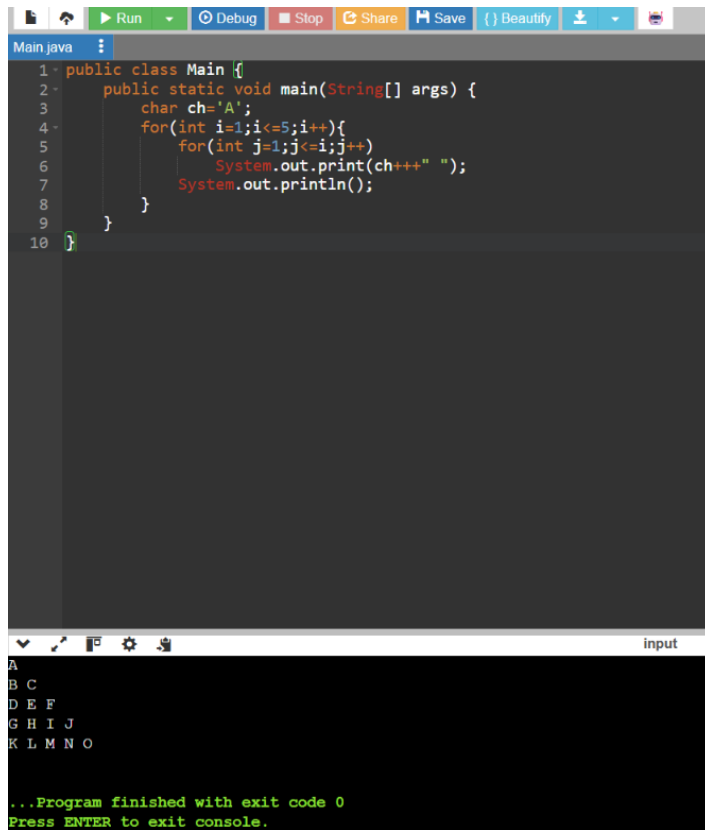


```
1 public class Main {
2     public static void main(String[] args) {
3         for(int i=1;i<=5;i++){
4             for(int j=1;j<=i;j++){
5                 if((j==1 || j==i || i==5) System.out.print("*");
6                 else System.out.print(" ");
7             }
8             System.out.println();
9         }
10 }
11 }
```

1
*
*
*
*

.. Program finished with exit code 0
Press ENTER to exit console.

16. Continuous Characters Triangle



The screenshot shows an IDE with a Java file named 'Main.java'. The code is as follows:

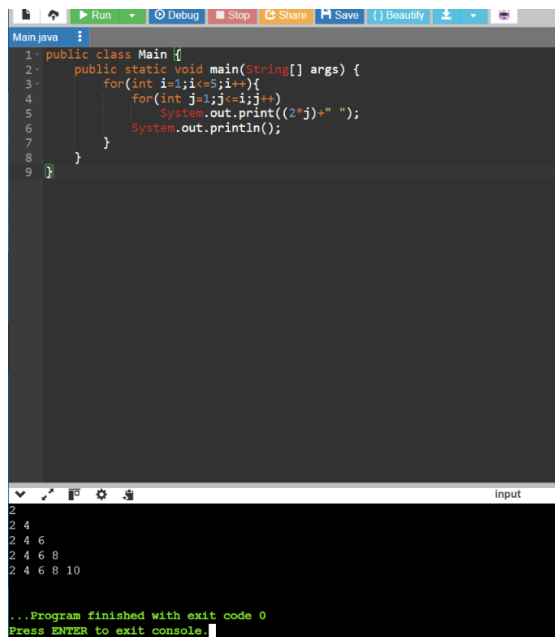
```
1 public class Main {  
2     public static void main(String[] args) {  
3         char ch='A';  
4         for(int i=1;i<=5;i++){  
5             for(int j=1;j<=i;j++){  
6                 System.out.print(ch+++" ");  
7             }  
8             System.out.println();  
9         }  
10    }
```

The output in the console is an alphabet triangle:

```
A  
B C  
D E F  
G H I J  
K L M N O
```

Below the output, it says: "...Program finished with exit code 0 Press ENTER to exit console."

17. Even Number Triangle



The screenshot shows an IDE with a Java file named 'Main.java'. The code is as follows:

```
1 public class Main {  
2     public static void main(String[] args) {  
3         for(int i=1;i<=5;i++){  
4             for(int j=1;j<=i;j++){  
5                 System.out.print((2*j)+" ");  
6             }  
7             System.out.println();  
8         }  
9     }
```

The output in the console is an even number triangle:

```
2  
2 4  
2 4 6  
2 4 6 8  
2 4 6 8 10
```

Below the output, it says: "...Program finished with exit code 0 Press ENTER to exit console."